

Task: “Mini Project Management App”

Objective

Build a simple full-stack app where users can manage projects and tasks.

Requirements

Frontend

1. Login / Signup Page

- Basic login and signup
- Mock authentication is acceptable
- JWT auth is a bonus

2. Dashboard Page

Show a list of projects with:

- Project name
- Project description
- Number of tasks
- Created date

3. Project Detail Page

Show all tasks related to a selected project.

4. Task Management

Users should be able to:

- Add task
- Edit task
- Delete task
- Change task status

Task statuses:

To Do

In Progress

Done

5. Search and Filter

- Search tasks by title
- Filter tasks by status

Backend

Create the following APIs:

POST /auth/register

POST /auth/login

GET /projects

POST /projects

GET /projects/:id

PUT /projects/:id

DELETE /projects/:id

POST /projects/:id/tasks

PUT /tasks/:id

DELETE /tasks/:id

Data Models

Project

```
{  
  "id": "unique-id",  
  "name": "Website Redesign",  
  "description": "Project description",  
  "createdAt": "date"  
}
```

Task

```
{  
  "id": "unique-id",  
  "projectId": "project-id",  
  "title": "Create landing page",  
  "description": "Task description",  
  "status": "todo | in-progress | done",  
  "assignedTo": "optional",  
  "createdAt": "date",  
  "updatedAt": "date"  
}
```

Preferred Tech Stack

Frontend

- React.js / Next.js
- Tailwind CSS / CSS Modules / plain CSS

Backend

- Node.js + Express.js
- MongoDB

Bonus

- JWT authentication
- Protected routes
- Form validation
- API validation
- Loading and empty states
- Deployment
- Basic tests
- Clean README